TP7

Exercise 1: Exporting sales to five destinations

Objective and source data

This exercise uses the `dbo.Ventes2024` table from TP6, stored in `TP6_Ventes` on the default SQL Server instance. The aim is to read the table once and export the same rows to CSV, Excel, Access, Oracle XE and PostgreSQL through a single SSIS Data Flow Task.

The source contains **three sales**:

Id	Annee	Region	Montant
1	2024	Tanger	1500.00
2	2024	Rabat	2300.00
30	2024	Tanger	500.00

The row with Id 30 was committed during the distributed-transaction test in TP6. It remains part of the source and is therefore included in this export. Each destination must receive **three rows**.

Package structure

The package `Sales_Export.dtsx` was created in Visual Studio with the SSIS extension. It contains one Data Flow Task named **Sales export**.

An **OLE DB Source** reads the four columns from `dbo.Ventes2024`, using the Microsoft OLE DB Driver for SQL Server and Windows Authentication. Its output feeds a **Multicast** transformation.

Multicast sends every input row to every connected output. This allows the five destinations to receive the complete dataset from one source read, without creating five separate source components.

CSV export

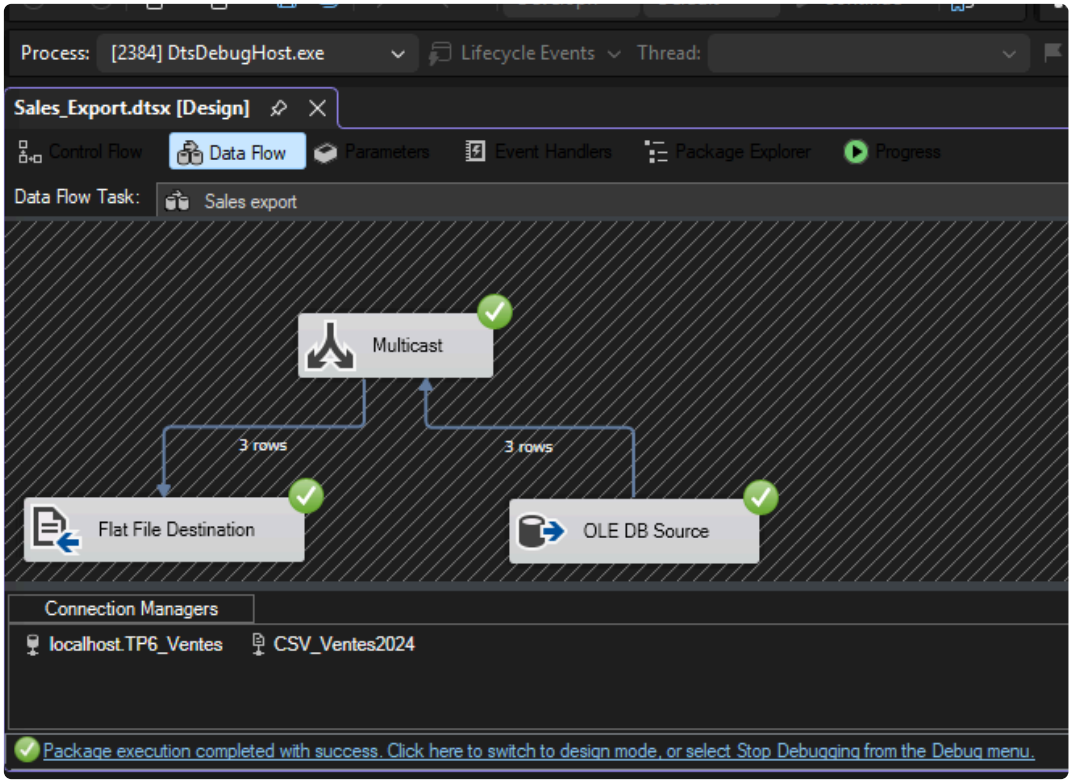
The first branch writes to `C:\TP7\ventes2024.csv` through a **Flat File Destination**.

The connection uses a delimited format with **semicolon separators**, Windows line endings and Unicode text. The header is written explicitly as:

```
Id;Annee;Region;Montant
```

All four input columns are mapped to the corresponding file columns. The destination is configured to overwrite the file on execution.

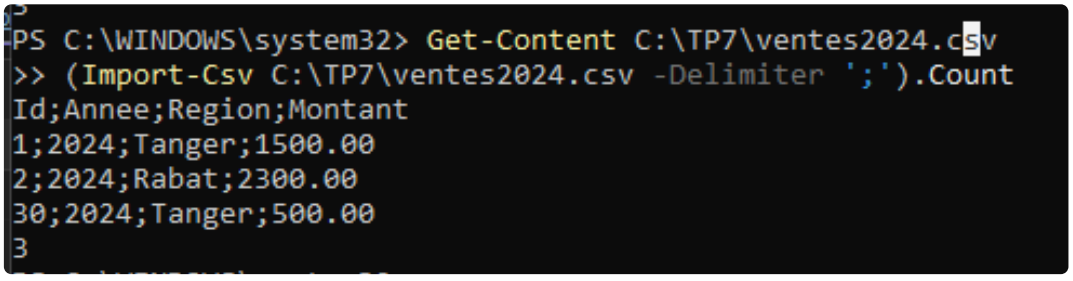
The initial branch test confirmed that three rows passed through the source and Multicast to the file destination.



The final file was checked in PowerShell with:

```
Get-Content C:\TP7\ventes2024.csv
(Import-Csv C:\TP7\ventes2024.csv -Delimiter ';').Count
```

The output contains one header line followed by the three sales. `Import-Csv` returns a count of **3**, excluding the header.



Excel, Access, Oracle and PostgreSQL destinations

Four additional outputs were connected to the same Multicast transformation.

Destination	Target	Connection
Excel	`C:\TP7\ventes2024.xlsx`	Excel connection manager using ACE

Destination	Target	Connection
Access	`SalesExport` in `C:\TP7\ventes.accdb`	ACE OLE DB
Oracle XE	`TP6.SALES_EXPORT` in `XEPDB1`	Oracle Provider for OLE DB
PostgreSQL	`public.sales_export` in `tp6_ventes`	64-bit PostgreSQL ODBC connection

The Excel destination writes the four fields to a worksheet with column names in the first row. The Access database contains an empty `SalesExport` table prepared with compatible integer, text and decimal fields.

Oracle and PostgreSQL use English destination column names. Their mappings preserve the meaning of the existing TP6 source fields:

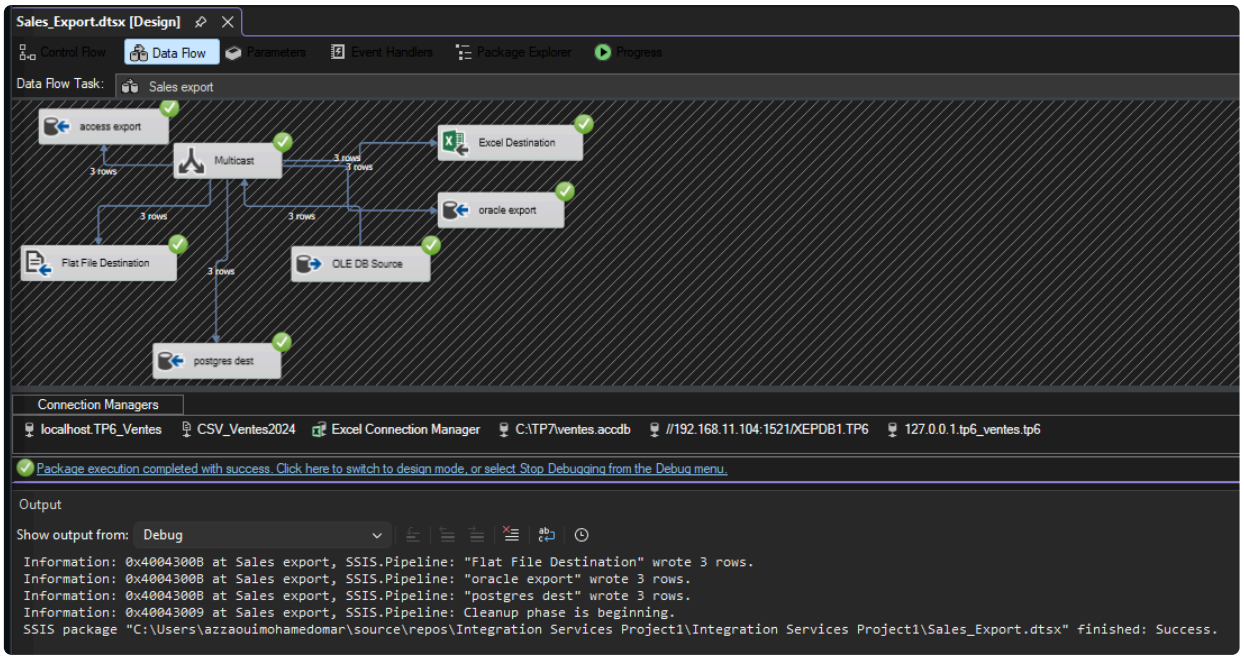
Source field	Oracle field	PostgreSQL field
Id	ID	id
Annee	SALE_YEAR	sale_year
Region	REGION	region
Montant	AMOUNT	amount

The Oracle connection uses the `TP6` account in `XEPDB1`. The PostgreSQL connection uses the `tp6` account and the database prepared in TP6.

Complete package execution

The complete package executed successfully. The source, Multicast and all five destinations show successful execution, with **three rows sent to each destination**.

The execution log also reports that the destinations wrote three rows and that the package finished with **Success**.



Verification of the exported data

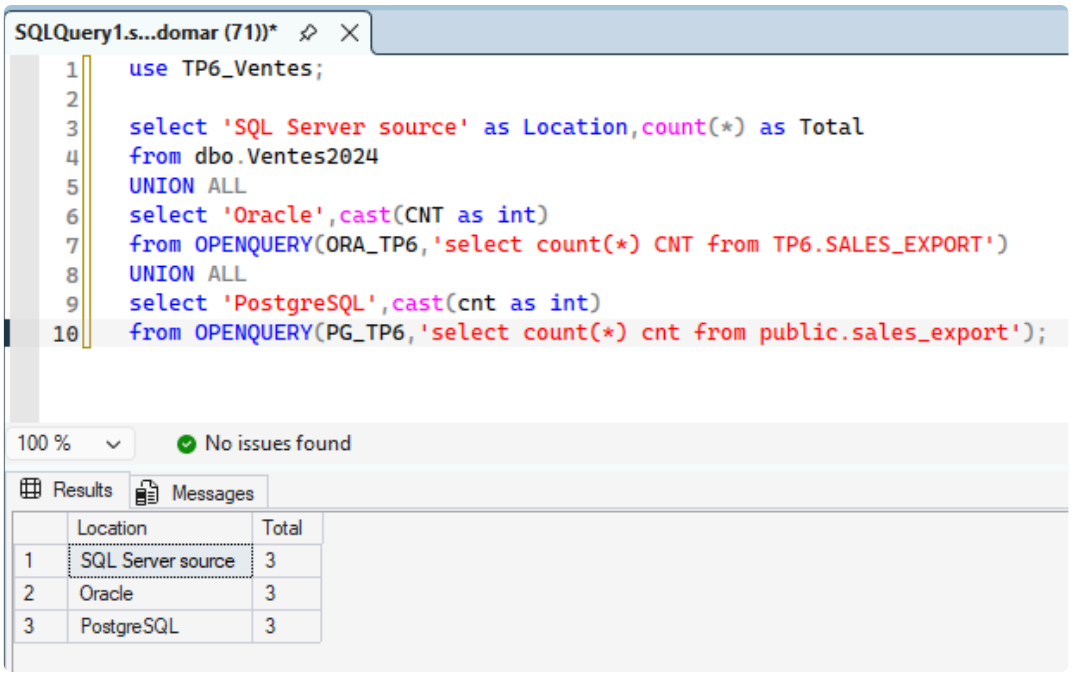
SQL Server, Oracle and PostgreSQL

A verification query was executed in SSMS on the default instance. It counts the local source rows and queries the Oracle and PostgreSQL destination tables through the existing linked servers.

The three results are identical:

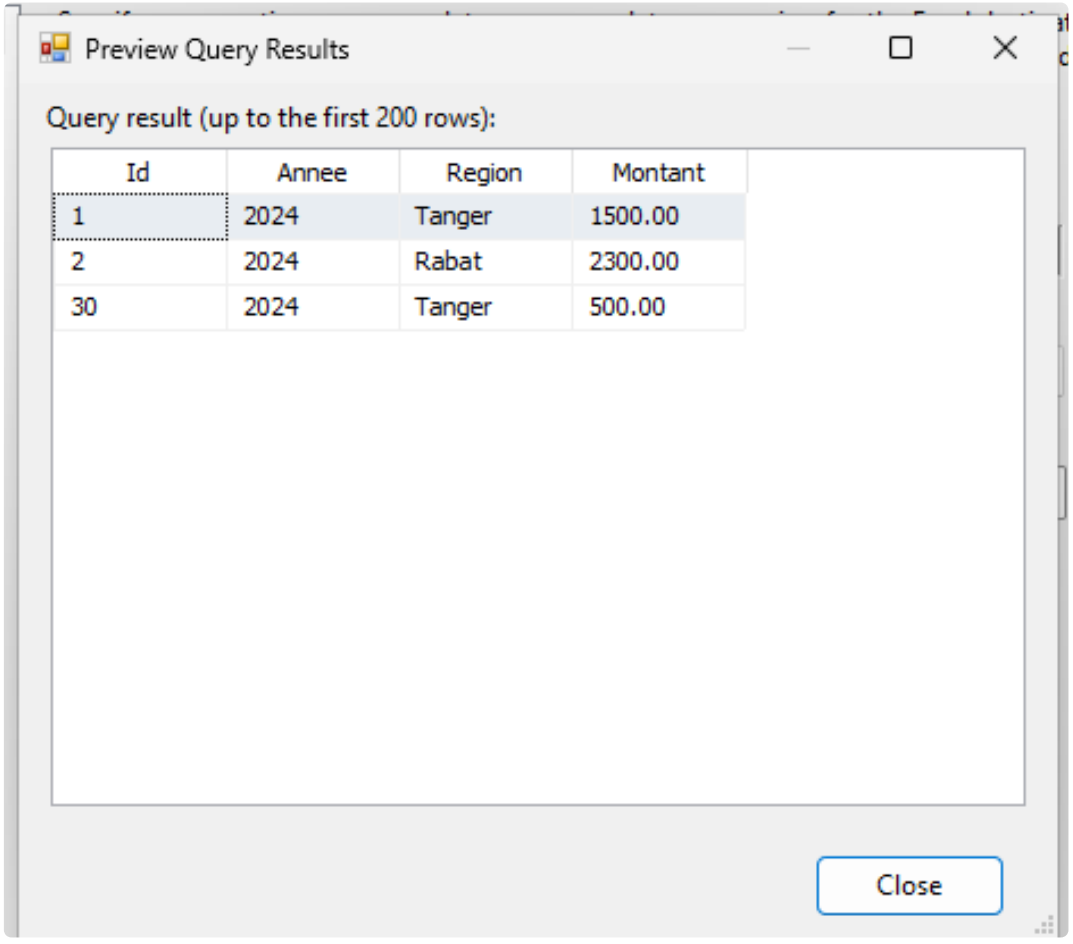
Location	Row count
SQL Server source	3
Oracle	3
PostgreSQL	3

This confirms that both database exports contain the same number of rows as the source.



Excel and Access

The Excel destination's **View Existing Data** option displays the three exported sales, with identifiers **1, 2 and 30** and their corresponding values.



The Access destination completed successfully and reported **three rows written**, as shown in the full Data Flow execution. The CSV content check, Excel preview and database count queries support the final verification of the export.

The package therefore distributes the three source sales to all five destinations through one Data Flow Task and a shared Multicast transformation.